

Администрирование локальных сетей

Лабораторная работа №12

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Цель

Приобретение практических навыков по настройке доступа локальной сети к внешней сети с использованием технологии NAT (Network Address Translation).

Выполнение лабораторной работы

Настройка маршрутизатора провайдера

```
Router>enable
Router#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host provider-aabdullakhi-gw-1
provider-aabdullakhi-gw-1(config)#line vty 0 4
provider-aabdullakhi-gw-1(config-line)#password cisco
provider-aabdullakhi-gw-1(config-line)#login
provider-aabdullakhi-gw-1(config-line)#exit
provider-aabdullakhi-gw-1(config)#
provider-aabdullakhi-gw-1(config)#line console 0
provider-aabdullakhi-gw-1(config-line)#password cisco
provider-aabdullakhi-gw-1(config-line)#login
provider-aabdullakhi-gw-1(config-line)#exit
provider-aabdullakhi-gw-1(config)#
provider-aabdullakhi-gw-1(config)#
provider-aabdullakhi-gw-1(config)#enable secret cisco
provider-aabdullakhi-gw-1(config)#service password-encryption
provider-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
provider-aabdullakhi-gw-1(config)#
provider-aabdullakhi-gw-1(config)#
```

Рис. 1: Настройка маршрутизатора провайдера

Настройка маршрутизатора провайдера

```
provider-aabdullakhi-gw-1(config)#int f0/0
provider-aabdullakhi-gw-1(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

provider-aabdullakhi-gw-1(config-if)#no shutdown

provider-aabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

provider-aabdullakhi-gw-1(config-if)#exit
provider-aabdullakhi-gw-1(config)#int f0/0.4
provider-aabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up

provider-aabdullakhi-gw-1(config-subif)#encapsulation dot1Q 4
provider-aabdullakhi-gw-1(config-subif)#ip address 198.51.100.1 255.255.255.240
provider-aabdullakhi-gw-1(config-subif)#description msk-donskaya
provider-aabdullakhi-gw-1(config-subif)#exit
provider-aabdullakhi-gw-1(config)#int f0/1
provider-aabdullakhi-gw-1(config-if)#no shutdown

provider-aabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

provider-aabdullakhi-gw-1(config-if)#ip address 192.0.2.1 255.255.255.0
provider-aabdullakhi-gw-1(config-if)#description internet
provider-aabdullakhi-gw-1(config-if)#exit
provider-aabdullakhi-gw-1(config)#exit
provider-aabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-aabdullakhi-gw-1#
```

Настройка коммутатора провайдера

```
Switch>
Switch>enable
Switch#con terminal
% Ambiguous command: "con terminal "
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host provider-aabdullakhi-sw-1
provider-aabdullakhi-sw-1(config)#line vty 0 4
provider-aabdullakhi-sw-1(config-line)#password cisco
provider-aabdullakhi-sw-1(config-line)#login
provider-aabdullakhi-sw-1(config-line)#exit
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#line console 0
provider-aabdullakhi-sw-1(config-line)#password cisco
provider-aabdullakhi-sw-1(config-line)#login
provider-aabdullakhi-sw-1(config-line)#exit
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#enable secret cisco
provider-aabdullakhi-sw-1(config)#service password-encryption
provider-aabdullakhi-sw-1(config)#username admin privilege 1 secret
cisco
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#
```

Copy

Paste

Настройка коммутатора провайдера

```
provider-aabdullakhi-sw-1(config)#int f0/1
provider-aabdullakhi-sw-1(config-if)#sw mode trunk

provider-aabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

provider-aabdullakhi-sw-1(config-if)#exit
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#int f0/24
provider-aabdullakhi-sw-1(config-if)#sw mode trunk

provider-aabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

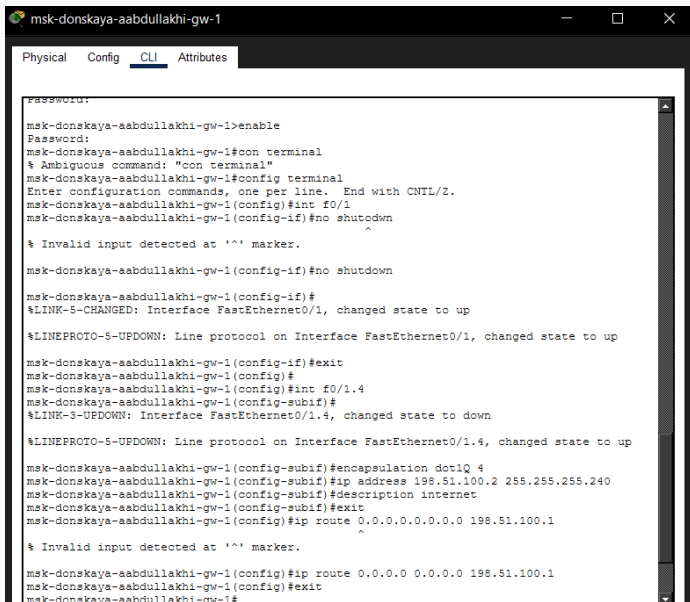
provider-aabdullakhi-sw-1(config-if)#exit
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#vlan 4
provider-aabdullakhi-sw-1(config-vlan)#name nat
provider-aabdullakhi-sw-1(config-vlan)#exit
provider-aabdullakhi-sw-1(config)#int vlan4
provider-aabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

provider-aabdullakhi-sw-1(config-if)#no shutdown
provider-aabdullakhi-sw-1(config-if)#exit
provider-aabdullakhi-sw-1(config)#
provider-aabdullakhi-sw-1(config)#
```

Рис. 4: Настройка коммутатора провайдера

Настройка интерфейса маршрутизатора сети «Донская»



The screenshot shows a terminal window titled 'msk-donskaya-aabdullakhi-gw-1' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying a series of configuration commands and system responses. The commands configure the router's enable password, enter configuration mode, set the terminal line speed to 19200 baud, and configure the FastEthernet0/1 and FastEthernet0/1.4 interfaces. The FastEthernet0/1 interface is configured with no shutdown, and the FastEthernet0/1.4 interface is configured with encapsulation dot1Q 4, IP address 198.51.100.2, and a description of 'internet'. The terminal output shows the state changes for these interfaces, including 'LINK-S-CHANGED' and 'LINEPROTO-S-UPDOWN'.

```
msk-donskaya-aabdullakhi-gw-1>enable
Password:
msk-donskaya-aabdullakhi-gw-1#con terminal
% Ambiguous command: "con terminal"
msk-donskaya-aabdullakhi-gw-1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-aabdullakhi-gw-1(config)#int f0/1
msk-donskaya-aabdullakhi-gw-1(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-gw-1(config-if)#no shutdown

msk-donskaya-aabdullakhi-gw-1(config-if)#
%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-if)#exit
msk-donskaya-aabdullakhi-gw-1(config)#
msk-donskaya-aabdullakhi-gw-1(config)#int f0/1.4
msk-donskaya-aabdullakhi-gw-1(config-subif)#
%LINK-S-UPDOWN: Interface FastEthernet0/1.4, changed state to down

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-subif)#encapsulation dot1Q 4
msk-donskaya-aabdullakhi-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
msk-donskaya-aabdullakhi-gw-1(config-subif)#description internet
msk-donskaya-aabdullakhi-gw-1(config-subif)#exit
msk-donskaya-aabdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-aabdullakhi-gw-1(config)#exit
msk-donskaya-aabdullakhi-gw-1#
```

Настройка NAT на маршрутизаторе «Донская»

```
msk-donskaya-aabdullakhi-gw-1(config)#ip nat pool main-pool 198.51.100.2 198.51.100.14
netmask 255.255.255.240
msk-donskaya-aabdullakhi-gw-1(config)#
```

Рис. 6: Настройка NAT на маршрутизаторе «Донская»

```
msk-donskaya-aabdullakhi-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#remark dk
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host
192.0.2.11 eq 80
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host
192.0.2.12 eq 80
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#remark departments
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host
192.0.2.13 eq 80
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#remark adm
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host
192.0.2.14 eq 80
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#remark admin
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#
msk-donskaya-aabdullakhi-gw-1(config-ext-nacl)#exit
msk-donskaya-aabdullakhi-gw-1(config)#
```

Рис. 7: Настройка NAT на маршрутизаторе «Донская»

Настройка статического NAT (проброс портов)

```
msk-donskaya-aabdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80
192.51.100.2 80
msk-donskaya-aabdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20
198.51.100.3 20
msk-donskaya-aabdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21
198.51.100.3 21
msk-donskaya-aabdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25
198.51.100.4 25
msk-donskaya-aabdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110
198.51.100.4 110
msk-donskaya-aabdullakhi-gw-1(config)#
```

Рис. 8: Настройка NAT на маршрутизаторе «Донская»

Проверка доступа из сети дисплейных классов

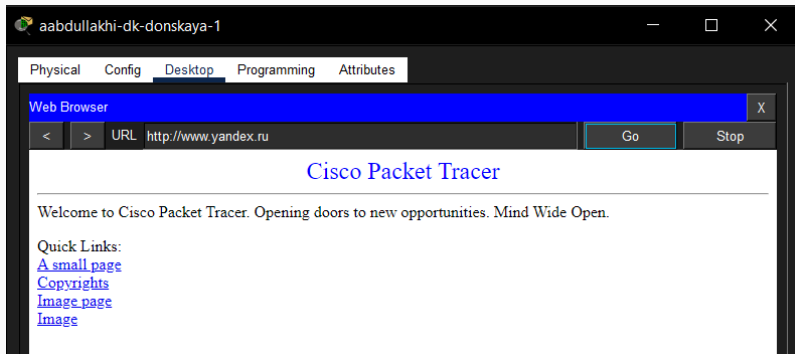


Рис. 9: Доступ к разрешённым ресурсам

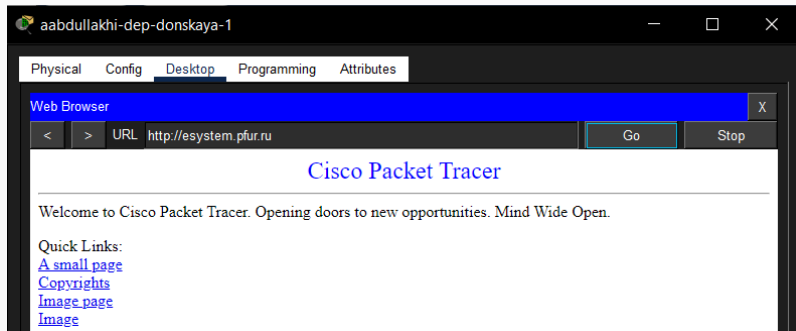


Рис. 10: Доступ к образовательному ресурсу из сети кафедр



Рис. 11: Доступ к сайту университета из сети администрации



Рис. 12: Отсутствие доступа у обычных пользователей

Проверка доступности серверов из
внешней сети

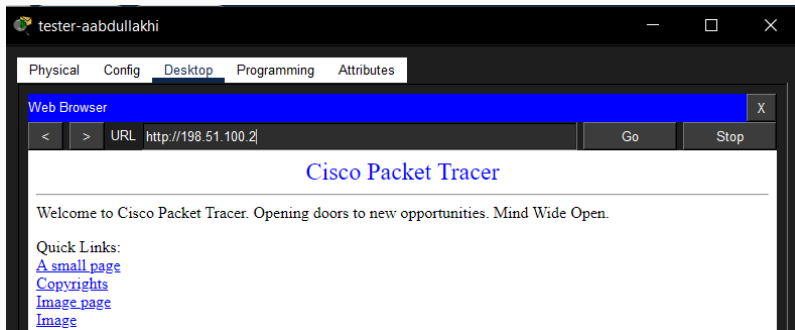
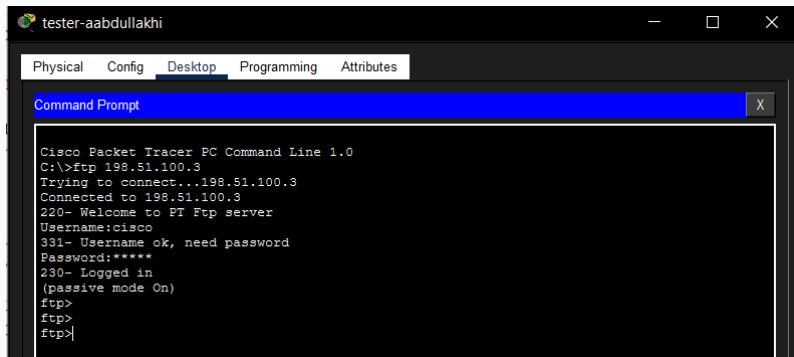


Рис. 13: Проверка WEB-сервера

Проверка FTP-сервера



The screenshot shows a Cisco Packet Tracer PC Command Line window titled "tester-aabdullakhi". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. The command prompt shows the following sequence of commands and responses:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 198.51.100.3
Trying to connect...198.51.100.3
Connected to 198.51.100.3
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
ftp>
ftp>|
```

Рис. 14: Проверка FTP-сервера

Проверка почтового сервера

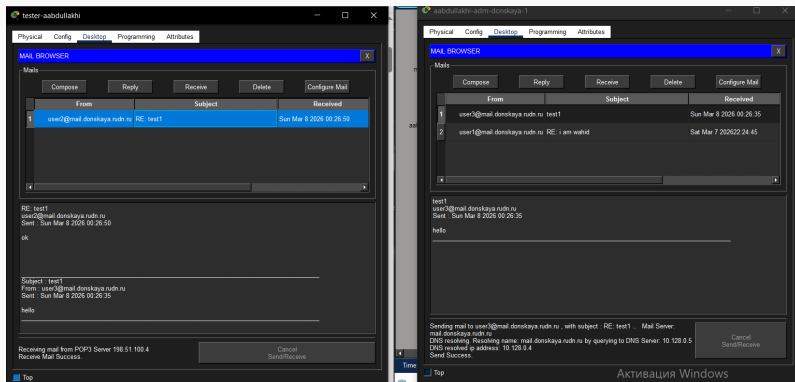


Рис. 15: Проверка почтового сервера

Вывод

В ходе выполнения лабораторной работы были освоены практические навыки настройки NAT на пограничном маршрутизаторе.

Реализованы:

- динамическая трансляция адресов для выхода в интернет;
- статический NAT для публикации внутренних серверов;
- разграничение доступа на основе ACL;
- проверка работоспособности WEB, FTP и почтовых сервисов.

Все цели работы достигнуты, конфигурация сохранена и протестирована.